SAMPSON BOATENG

Data Analyst | (207) 332-5395 | samboateng190@gmail.com | github.com/Boatengs | linkedin.com/in/sam-boateng

PROFESSIONAL SUMMARY

Data analyst with an M.S. in Applied Machine Intelligence and a B.S. in Computer Science, experienced in transforming financial, donor, healthcare, and operational data into reliable analysis and decision-ready insights. Proficient in SQL, Excel, Python, Tableau, and Power BI, with strengths in data cleaning, validation, exploratory analysis, visualization, forecasting, and dashboard development. Communicates technical findings clearly to business stakeholders and maintains rigorous standards for data accuracy and documentation.

EDUCATION

Northeastern University, The Roux Institute | Portland, ME — June 2025
Master of Science, Applied Machine Intelligence

Saratov State University | Saratov, Russia — June 2023
Bachelor of Science, Computer Science

TECHNICAL SKILLS

Analytics & Visualization: SQL, Excel, Tableau, Power BI, Python (Pandas, NumPy), Matplotlib, Seaborn
Statistical Analysis: Regression, clustering, time-series forecasting, anomaly detection, hypothesis testing
Databases & Big Data: SQL, Apache Spark, Hadoop, Hive
Tools & Workflow: Git, Jira, Dataiku, Microsoft Office Suite

WORK EXPERIENCE

Finance, Development & IT Administration Support Analyst
Village Health Works | New York, NY — June 2025 - Present

- Analyze financial and donor datasets, validate records, investigate discrepancies, and explain budget-to-actual variances.
- Prepare reconciliations, management reports, and forecasts that support planning and executive decision-making.
- Maintain donor data and conduct exploratory analysis of giving patterns to inform fundraising strategy and stakeholder outreach.
- Establish documentation and data-quality standards that improve consistency, usability, and confidence in organizational records.
- Support business systems, troubleshoot data and tool issues, and onboard staff to shared technology workflows.

Programmer Intern
Bosch Techno-Engineering Ltd | Saratov, Russia — 2021 - 2023

- Supported software development and quality-assurance cycles by writing and executing test cases to verify system behavior.
- Diagnosed and resolved defects through structured debugging, contributing to a 20% improvement in system stability.
- Developed an internal encryption and decryption testing utility that strengthened validation and security practices.

SELECTED DATA & MACHINE LEARNING PROJECTS

Full code and technical documentation: github.com/Boatengs

- **Price Elasticity Modeling:** Built demand and pricing models with Random Forest, XGBoost, and ARIMA; delivered results through a Plotly Dash dashboard.
- **Water Quality Analysis:** Applied Random Forest, K-Means, and Isolation Forest with Pandas and SciPy to detect environmental anomalies.
- **Healthcare Resource Modeling:** Developed and compared Logistic Regression, Random Forest, SVM, and PCA-based models in Scikit-learn.
- **Sentiment Analyzer:** Fine-tuned DistilBERT on 25,000 IMDb reviews, achieved 91.2% accuracy, and deployed the model with Gradio.
- **Sports Q&A Chatbot:** Built a retrieval-augmented application using LangChain, ChromaDB, Llama 3.2, and a FastAPI backend.
- **Medical Q&A Model:** Fine-tuned Mistral 7B with LoRA/QLoRA using PEFT and TRL; published the trained adapter to Hugging Face Hub.
- **Skin Disease Classifier:** Applied EfficientNetB0 transfer learning with Grad-CAM explanations and deployed the classifier through Gradio.